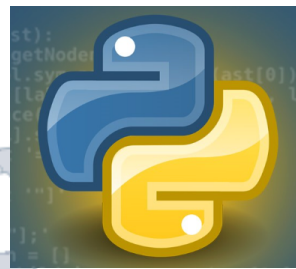


Analysis of Real-World Attack Activity Against an Azure Windows Honeypot

A Practical SOC Project for Log Analysis, Threat Detection, and Attack Visualisation in Microsoft Azure



*Prepared By
Sivateja Mattaparthi*

Master of Information Technology

June 2026

Montréal (CA)
1.78k

Gazipur (BD)
590

Amsterdam (NL)
235

San Jose (US)
15

Other
7

Casablanca (MA)
6

1) Project Overview

The purpose of this project was to build a cloud-hosted Windows system capable of attracting, recording, and analysing real-world cyber attack activity. A Windows virtual machine was deployed in Microsoft Azure and intentionally exposed to the public internet to observe how attackers interact with an accessible system.

As malicious actors attempted to gain unauthorised access, their activities were captured within Windows Security Event Logs. These logs were collected and centralised using Azure Log Analytics and subsequently analysed through Microsoft Sentinel, Microsoft's cloud-native Security Information and Event Management (SIEM) platform.

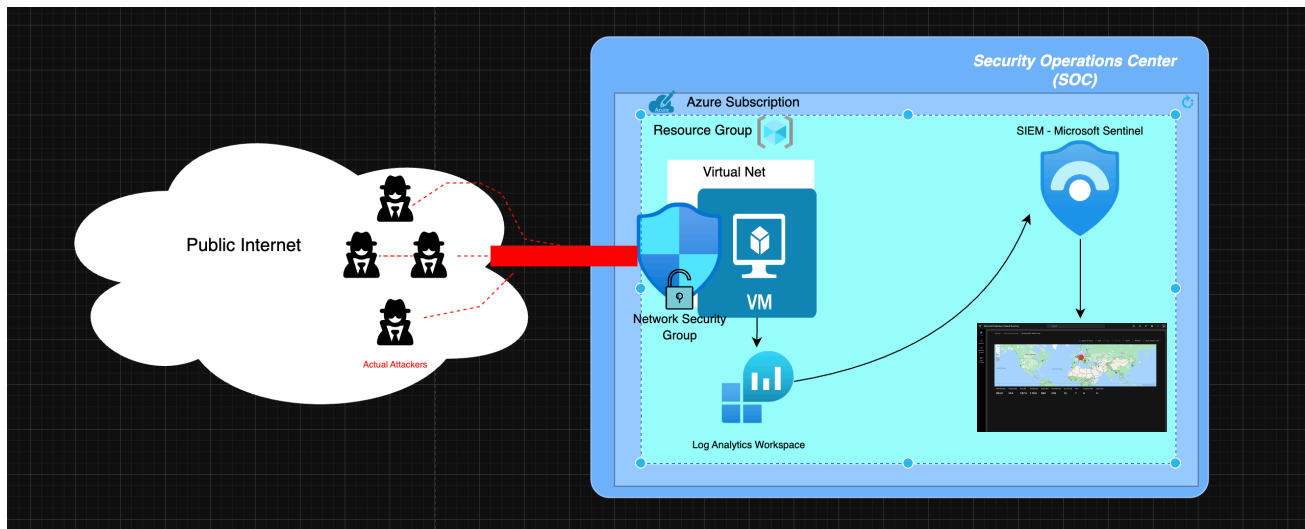
To gain greater visibility into the origin of attack activity, attacker IP addresses were extracted from failed authentication events and enriched with geolocation and network ownership information using Python and the IPinfo API. The enriched data was then integrated into Microsoft Sentinel and visualised through an interactive global attack map, enabling the identification of attack patterns and geographic trends.

This project demonstrates practical experience in cloud infrastructure deployment, security monitoring, log management, threat intelligence enrichment, security analytics using Kusto Query Language (KQL), automation with Python, and Security Operations Center (SOC) investigation workflows. It also highlights how modern security teams collect, enrich, and analyse security telemetry to better understand emerging threats and attacker behaviour.

Project Objectives:

- Deploy a Windows virtual machine in Microsoft Azure.
- Expose the system to the public internet to collect real-world attack activity.
- Collect and centralise Windows Security Event Logs using Azure Log Analytics.
- Monitor and investigate security events using Microsoft Sentinel.
- Analyse failed authentication attempts using Kusto Query Language (KQL).
- Extract and enrich attacker IP addresses with geolocation and network information.
- Automate threat intelligence enrichment using Python and the IPinfo API.
- Create a custom Sentinel Watchlist for attack attribution.
- Visualise attack activity through geospatial dashboards and maps.
- Gain hands-on experience with SOC monitoring, threat hunting, and incident investigation workflows.

2) System Architecture Diagram



The architecture was designed to simulate a publicly accessible Windows system and monitor real-world attack activity using Microsoft Sentinel.

A Windows Virtual Machine (VM) was deployed within a Microsoft Azure Virtual Network (VNet) and protected by a Network Security Group (NSG). To intentionally attract attack traffic and generate security events, the default security posture of the environment was deliberately weakened.

At the network layer, the Network Security Group's inbound filtering rules were modified to allow traffic from **any source to any destination**, effectively exposing the virtual machine to the public internet. Additionally, the Windows Defender Firewall within the virtual machine was disabled, reducing host-based network protections and allowing unsolicited traffic to reach the system.

Once exposed, automated scanning tools and threat actors from across the internet began attempting to access the virtual machine. These activities generated Windows Security Events, particularly failed authentication events (Event ID 4625), which were recorded by the operating system.

The collected security logs were forwarded to an Azure Log Analytics Workspace, where they were centralised and stored for analysis. Microsoft Sentinel was connected to the Log Analytics Workspace and served as the Security Information and Event Management (SIEM) platform for monitoring, investigation, and threat analysis.

Using Kusto Query Language (KQL), failed authentication attempts were extracted and analysed. Attacker IP addresses were subsequently enriched with geolocation and network ownership information using external threat intelligence data. The enriched information was then visualised through interactive dashboards and geographic attack maps within Microsoft Sentinel, enabling the identification of attack sources and attack patterns across different regions of the world.

Component	Purpose
Public Internet	Source of real-world attack traffic and automated scanning activity
Network Security Group (NSG)	Network-level filtering mechanism intentionally weakened by modifying inbound rules to allow unrestricted access
Windows Virtual Machine	Target system used to collect attack telemetry
Windows Defender Firewall	Host-based firewall intentionally disabled to reduce network restrictions
Azure Virtual Network (VNet)	Provides networking infrastructure for the virtual machine
Log Analytics Workspace	Centralised collection and storage of Windows Security Logs
Microsoft Sentinel	SIEM platform used for monitoring, querying, investigation, and visualization
KQL Queries	Used to identify failed authentication attempts and attacker activity
GeoIP Enrichment	Adds geographic and network ownership context to attacker IP addresses
Attack Map Dashboard	Visual representation of global attack sources and attack frequency

Security Note:

The intentionally weakened configuration used in this project was created solely for educational and research purposes to observe real-world attack behaviour. Such configurations should never be used in production environments, where strict Network Security Group rules, host-based firewalls, strong authentication mechanisms, and least-privilege access controls are essential security requirements.

3) Azure Environment Setup

Environment Deployment

To create a realistic cloud-based monitoring environment, a Windows virtual machine was deployed within Microsoft Azure. The virtual machine served as the primary target system used to collect and analyse real-world attack activity originating from the public internet.

The deployment was performed within a dedicated Azure Resource Group, allowing all project resources to be logically organised and managed from a single location. The environment consisted of networking, compute, logging, and security monitoring components that worked together to collect and analyse security telemetry.

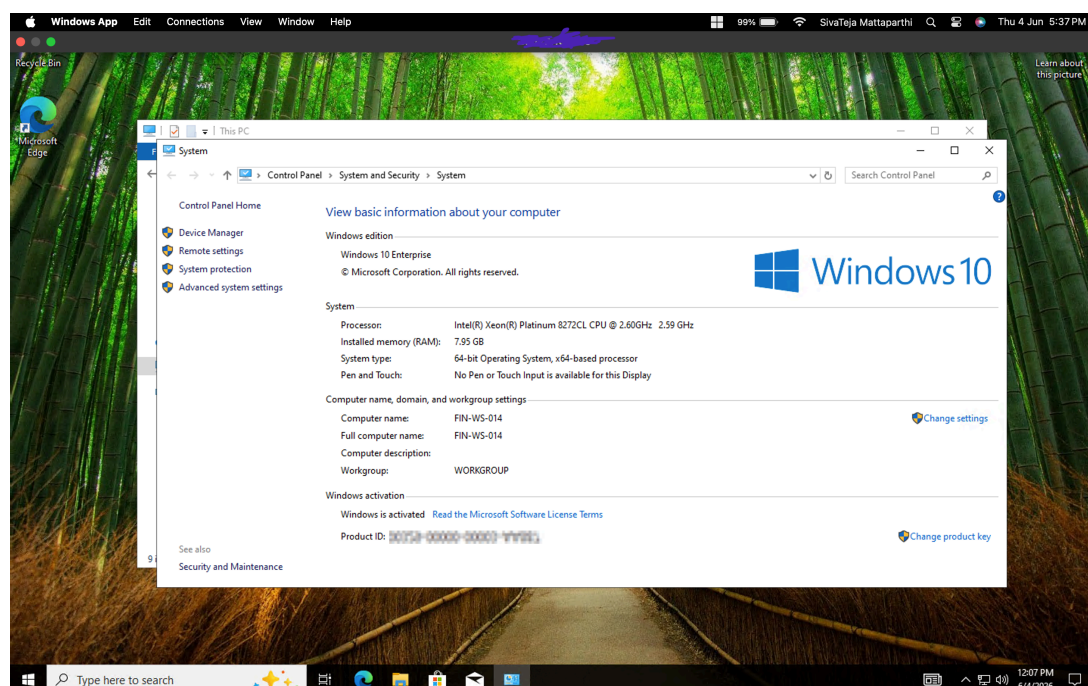
The Windows virtual machine was assigned a public IP address, enabling external connectivity from the internet. This allowed automated scanners and threat actors to discover and interact with the system, generating security events for analysis.

Resources Deployed

Resource	Purpose
Resource Group	Logical container used to manage all project resources
Virtual Machine (Windows)	Target system used to generate security telemetry
Virtual Network (VNet)	Provides networking infrastructure for the VM
Network Security Group (NSG)	Controls inbound and outbound network traffic
Public IP Address	Allows internet connectivity to the virtual machine
Log Analytics Workspace	Centralised log collection and storage
Microsoft Sentinel	Security monitoring and threat analysis platform

Virtual Machine Configuration

Configuration Item	Value
Operating System	Windows 10 Enterprise 22H2
Deployment Platform	Microsoft Azure
Public Accessibility	Enabled
Authentication Method	Username and Password
Logging Source	Windows Security Event Logs



The Virtual machine can be accessed by:

Mac: Through Windows app available in Appstore (connect to VM using the Public IP and user credentials)

Windows: Through Remote Desktop (Microsoft Remote Desktop)

Network Security Group Configuration

To attract and observe real-world attack activity, the virtual machine was intentionally made accessible from the public internet.

The Network Security Group (NSG) attached to the virtual machine was modified to allow inbound traffic from any source. This permissive configuration reduced network-level restrictions and increased the likelihood of receiving attack traffic from automated scanners and malicious actors operating across the internet.

In addition, the Windows Defender Firewall running within the virtual machine was disabled to further reduce host-level filtering and ensure attack activity would be recorded within the Windows Security Event Logs.

This intentionally weakened security posture was created solely for research and educational purposes and enabled the collection of realistic attack telemetry for subsequent analysis.

The screenshot displays the Microsoft Azure portal interface for configuring a Network Security Group (NSG). The main pane shows the 'FIN-WS-014-nsg' resource, which is a Network security group. The 'Overview' tab is selected, showing details like location (East US 2), subscription (Azure subscription 1), and associated resources (0 subnets, 1 network interfaces). Below this, a table lists the security rules. The 'Inbound Security Rules' section is expanded, showing a rule named 'Caution_Allowanyinboundrule' with a priority of 100. This rule is configured to allow inbound traffic from any source to any destination on any port. The right-hand pane shows the detailed configuration for this rule, including source, destination, service, and action settings. The rule is configured to allow inbound traffic from any source to any destination on any port. The right-hand pane also displays several security alerts, such as 'RDP port 3389 is exposed to the Internet' and 'SSH port 22 is exposed to the Internet'.

Priority	Name	Port	Protocol	Source	Destination	Action
100	Caution_Allowanyinboundrule	Any	Any	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Fig: A new inbound rule attached which allows internet traffic from any source widely opening the NSG

RG_SOC_L1
Resource group

✦ ☆ ...

✕

+ Create
Manage view
Delete resource group
Refresh
Export to CSV
Group by none

Overview
Activity log
Access control (IAM)
Tags
Resource visualizer
Events
Settings
Cost Management
Monitoring
Automation
Help

Essentials
JSON View

Resources
Recommendations

Filter for any field...
Type equals all
Location equals all
Add filter

☐	Name ↑	Type	Location
☐	00224228-a791-4130-8666-e32822dbd4f7 (Window)	Azure Workbook	East US 2
☐	DataCR-SOC-L1	Data collection rule	East US 2
☐	FIN-WS-014	Virtual machine	East US 2
☐	FIN-WS-014-ip	Public IP address	East US 2
☐	FIN-WS-014-nsg	Network security group	East US 2
☐	fin-ws-014609	Network Interface	East US 2
☐	FIN-WS-014_OsDisk_1_de42e1e2fa9a4c608bc76eec8	Disk	East US 2
☐	LAW-SOC-L1	Log Analytics workspace	East US 2
☐	SecurityInsights(law-soc-l1)	Solution	East US 2
☐	VN_SOC_L1	Virtual network	East US 2

Fig: Components of the Resource Group

Deployment Outcome

Following deployment and exposure to the internet, the virtual machine immediately began receiving unsolicited connection attempts from multiple geographic locations. These events were successfully captured within Windows Security Logs and forwarded to Azure Log Analytics for further analysis using Microsoft Sentinel.

4) Log Collection and Microsoft Sentinel Integration

Log Analytics Workspace Deployment

To collect and analyse security telemetry generated by the Windows virtual machine, an Azure Log Analytics Workspace was first created. The workspace serves as a centralised repository for storing, querying, and investigating log data collected from connected resources.

By centralising log data within a Log Analytics Workspace, security events generated by the virtual machine could be retained, searched, and analysed using KQL.

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with various service categories. The main area shows the 'LAW-SOC-L1' Log Analytics workspace. A 'New Query' window is open, showing a KQL query that filters for failed login events (EventID == 4625) and projects specific fields. The query results are displayed in a table with columns: TimeGenerated [UTC], Account, Computer, EventID, Activity, and IpAddress. The table shows 12 entries of failed logins from 5/6/2026.

TimeGenerated [UTC]	Account	Computer	EventID	Activity	IpAddress
> 5/6/2026, 9:59:19.901 am	\REMOTE	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.728 am	\USER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.558 am	\ADMIN	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.386 am	\ADMINISTRATOR	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.211 am	\SERVER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.047 am	\REMOTE	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.878 am	\USER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.709 am	\ADMIN	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.539 am	\ADMINISTRATOR	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.373 am	\SERVER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.201 am	\REMOTE	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16

Fig 4.1: Log Analytics Workspace showing query Results using KQL of occurred failed login events.

Microsoft Sentinel Deployment

After the Log Analytics Workspace was created, Microsoft Sentinel was deployed and connected to the workspace. During the deployment process, the newly created Log Analytics Workspace was selected as the data source for Microsoft Sentinel.

This integration enabled Sentinel to function as the Security Information and Event Management (SIEM) platform for the environment, providing capabilities for log collection, monitoring, investigation, threat hunting, and visualisation.

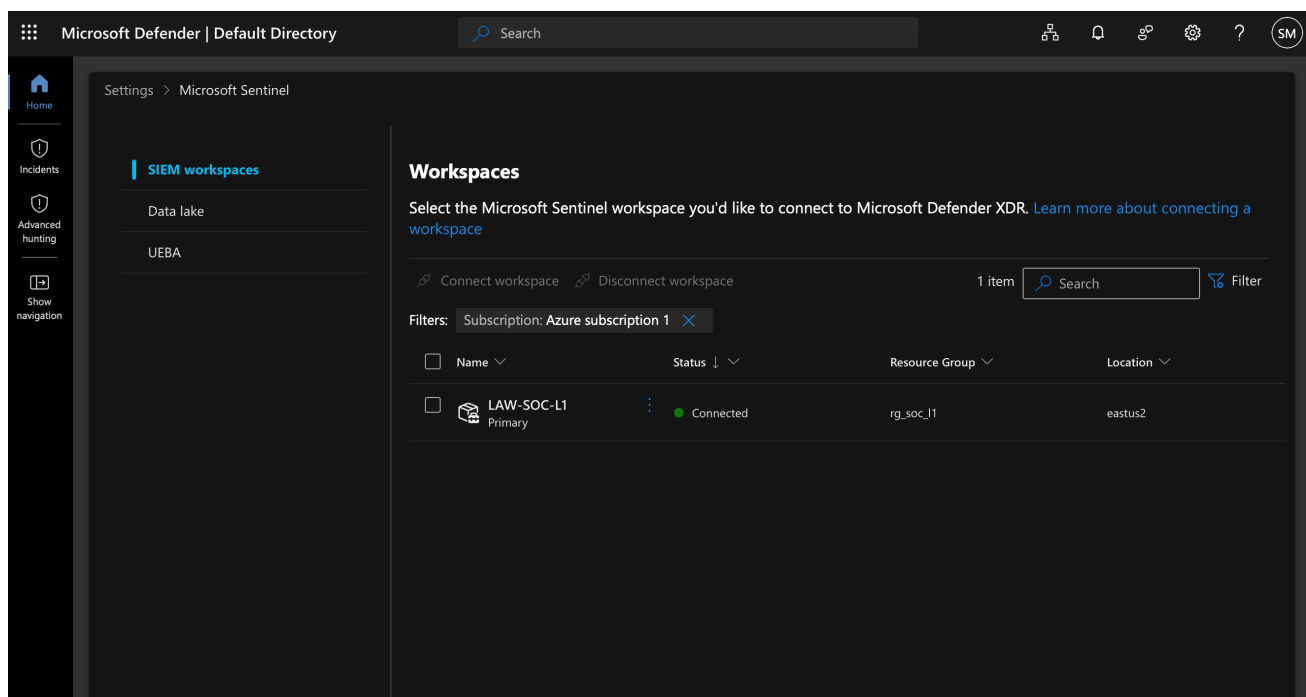


Fig 4.2: Log Analytics Workspace added in Sentinel within Microsoft Defender platform.

Security Event Collection Configuration

To begin collecting Windows Security Events from the virtual machine, the “Configuration Hub” within Microsoft Sentinel was used.

The following steps were performed:

1. Navigate to Microsoft Sentinel → Configuration → Configuration Hub.
2. Search for and install the Windows Security Events.
3. Configure the data connector to collect security events from the Windows virtual machine.
4. Install the required Azure monitoring extensions on the virtual machine.
5. Verify successful data ingestion into the Log Analytics Workspace.

Once the required components and extensions were installed, Windows Security Event Logs began flowing from the virtual machine into the Log Analytics Workspace.

Verification of Data Ingestion

To verify successful log collection, KQL queries were executed within Microsoft Sentinel and Log Analytics.

The following query was used to retrieve failed authentication events:

SecurityEvent

| where EventID == 4625

Successful execution of the query confirmed that Windows Security Event Logs were being collected, forwarded to the Log Analytics Workspace, and made available for investigation through Microsoft Sentinel.

These events later served as the primary data source for attacker IP extraction, geolocation enrichment, and attack visualisation activities performed during the project.

The screenshot displays the Microsoft Sentinel KQL query editor and results. The query is: `SecurityEvent | where EventID == 4625`. The results table shows 10 rows of failed authentication events.

TimeGenerated [UTC]	Account	AccountType	Computer
> 5/6/2026, 9:41:44.010 am	\REMOTE	User	FIN-WS-C
> 5/6/2026, 9:41:43.841 am	\USER	User	FIN-WS-C
> 5/6/2026, 9:41:43.668 am	\ADMIN	User	FIN-WS-C
> 5/6/2026, 9:41:43.502 am	\ADMINISTRATOR	User	FIN-WS-C
> 5/6/2026, 9:41:43.334 am	\SERVER	User	FIN-WS-C
> 5/6/2026, 9:41:43.165 am	\REMOTE	User	FIN-WS-C
> 5/6/2026, 9:41:42.995 am	\USER	User	FIN-WS-C
> 5/6/2026, 9:41:42.814 am	\ADMIN	User	FIN-WS-C
> 5/6/2026, 9:41:42.724 am	\ADMINISTRATOR	User	FIN-WS-C

3s 999ms | Display time (UTC+00:00) | Query details | 1 - 10 of 15000

5) Kusto Query Language (KQL) Log Analysis

After confirming that Windows Security Event Logs were successfully being ingested into Microsoft Sentinel, Kusto Query Language (KQL) was used to investigate authentication activity targeting the internet-exposed virtual machine.

KQL is Microsoft's query language used throughout Azure Monitor, Log Analytics, and Microsoft Sentinel for searching, filtering, correlating, and analysing security data.

The primary objective of this analysis was to identify failed authentication attempts, determine the most active attacker IP addresses, and extract data for further threat intelligence enrichment.

Extracting Relevant Security Information

To better understand the attack activity, relevant fields were selected from the event logs. The following query was used:

SecurityEvent

| where EventID == 4625

| project TimeGenerated, Account, Computer, EventID, Activity, IPAddress

This query provided key information including:

- Timestamp of the attack attempt
- Target username
- Target system
- Event type
- Source IP address

By examining these fields, attacker behaviour and login patterns could be investigated in greater detail.

Safari File Edit View History Bookmarks Window Help portal.azure.com

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Log Analytics workspaces > LAW-SOC-L1

LAW-SOC-L1 | Logs Log Analytics workspace

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Logs
Resource visualizer
Settings
Classic
Monitoring
Automation
Help

New Query 1* ... +

Run | Time range: Last 24 hours Show: 15000 results KQL mode

```
1 SecurityEvent
2 | where EventID == 4625
3 | project TimeGenerated, Account, Computer, EventID, Activity, IPAddress
```

Results Chart

TimeGenerated [UTC]	Account	Computer	EventID	Activity	IpAddress
> 5/6/2026, 9:59:19.901 am	\REMOTE	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.728 am	\USER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.558 am	\ADMIN	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.386 am	\ADMINISTRATOR	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.211 am	\SERVER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:19.047 am	\REMOTE	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.878 am	\USER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.709 am	\ADMIN	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.539 am	\ADMINISTRATOR	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.373 am	\SERVER	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16
> 5/6/2026, 9:59:18.201 am	\REMOTE	FIN-WS-014	4625	4625 - An account failed to log on.	148.251.140.16

3s 837ms Display time (UTC+00:00) Query details 1 - 12 of 15000

Identifying High-Volume Attack Sources

To determine which IP addresses were responsible for the greatest number of failed authentication attempts, aggregation techniques were applied. The following query was used:

SecurityEvent

| where EventID == 4625

| summarize Attempts=count() by IpAddress

| order by Attempts desc

This query grouped events by source IP address and calculated the number of authentication failures generated by each host.

The results revealed several highly active sources generating thousands of failed authentication attempts against the virtual machine.

The screenshot displays the Microsoft Azure portal interface. On the left is a navigation pane with options like 'Create a resource', 'Home', 'Dashboard', and 'All services'. The main area shows the 'LAW-SOC-L1 | Logs' section. A KQL query is entered in the 'New Query 1' editor:

```
1 SecurityEvent
2 | where EventID == 4625
3 | summarize Attempts=count() by IPAddress, Account
4 | order by Attempts desc
```

The query results are displayed in a table with columns 'IpAddress', 'Account', and 'Attempts'. The results show 10 entries, each with an IP address, an account name, and a count of 1 attempt.

IpAddress	Account	Attempts
> 160.177.168.218	\AZUREADMIN	1
> 160.177.168.218	\LABADMIN	1
> 45.142.193.166	\administrator	1
> 34.77.166.77	\	1
> 80.66.83.80	\Test	1
> 4.158.78.129	\LABADMIN	1
> 37.255.239.81	\ADMINISTRATOR	1
> 217.154.98.229	\LOCALADMIN	1
> 217.154.98.229	\AZUREUSER	1
> 217.154.98.229	\STUDENT	1
> 46.105.132.55	WORKGROUP\GUEST	1

Identifying Usernames Targeted by Attackers

To investigate which usernames were being targeted during authentication attempts, failed logon events were grouped by source IP address and account name.

SecurityEvent

| where EventID == 4625

| summarize Attempts=count() by IPAddress, Account

| order by Attempts desc

This analysis helped identify common usernames targeted by automated password guessing and brute-force attacks.

The results demonstrated how attackers repeatedly attempted authentication using various account names in an effort to gain unauthorised access.

The screenshot displays the Microsoft Azure portal interface. On the left is the navigation pane with options like 'Create a resource', 'Home', 'Dashboard', and 'Log Analytics workspaces'. The main area shows the 'LAW-SOC-L1 | Logs' workspace. A KQL query is entered in the 'New Query 1' editor:

```
1 SecurityEvent
2 | where EventID == 4625
3 | summarize Attempts=count() by IPAddress, Account
4 | order by Attempts desc
```

The query results are displayed in a table with columns 'IpAddress', 'Account', and 'Attempts'. The table shows 16 rows of data, sorted by the number of attempts in descending order.

IpAddress	Account	Attempts
> 148.251.140.16	\ADMIN	28934
> 148.251.140.16	\SUPPORT	17772
> 148.251.140.16	\TEST	17772
> 220.86.177.133	\ADMINISTRATOR	7865
> 80.94.95.83	\ADMIN	2328
> 80.94.95.83	\ADMINISTRADOR	2327
> 80.94.95.83	\BACKUP	2327
> 80.94.95.83	\USER	2327
> 80.94.95.83	\ADMINISTRATOR	2327
> 80.94.95.83	\SYSTEM	2326
> 51.79.47.237	\ADMINISTRATOR	1781

At the bottom of the results, it indicates '1s 849ms' and 'Display time (UTC+00:00)'. The status bar shows 'Query details' and '5 - 16 of 5086'.

Exporting Attacker Intelligence

After identifying the most active attacker IP addresses, the results were exported as a CSV file for external enrichment.

The exported dataset contained:

- Source IP Address
- Number of Failed Attempts
- Target Account Information

6) Threat Intelligence Enrichment Using Python and the IPinfo API

While the security logs provided information about failed authentication attempts, they did not provide contextual details such as the geographic location or network ownership of the source IP addresses. To address this limitation, external threat intelligence enrichment was performed using the IPinfo API and a custom Python automation script.

The objective of this process was to transform raw IP addresses into actionable intelligence by identifying the approximate geographic origin and associated network provider for each attacker.

IPinfo API Integration

To obtain geolocation information, an account was created on the IPinfo platform and an API access token was generated.

The API token was used to authenticate requests from the Python script, allowing additional information to be retrieved for each attacker IP address.

The enrichment process provided following attributes:

- Network Provider (ASN/Organisation)
- City
- Country
- Latitude
- Longitude

These attributes were later used to create geographic visualisations within Microsoft Sentinel.

Python in Enrichment process

A custom Python script was developed to automate the enrichment process.

The script performed the following actions:

1. Read the exported CSV file containing attacker IP addresses.
2. Query the IPinfo API for each IP address.
3. Retrieve geolocation and network ownership information.
4. Append the enrichment data to the original dataset.
5. Generate a new enriched CSV file for ingestion into Microsoft Sentinel.

Enriched Data Creation

After processing the exported attacker data, a new dataset was generated containing both the original attack statistics and the newly enriched intelligence information.

The dataset includes attributes:

Field	Description
IP Address	Source attacker IP
Attempts	Number of failed logon attempts
Network	Network provider / ASN
City	Approximate city location
Country	Country of origin
Latitude	Geographic latitude
Longitude	Geographic longitude

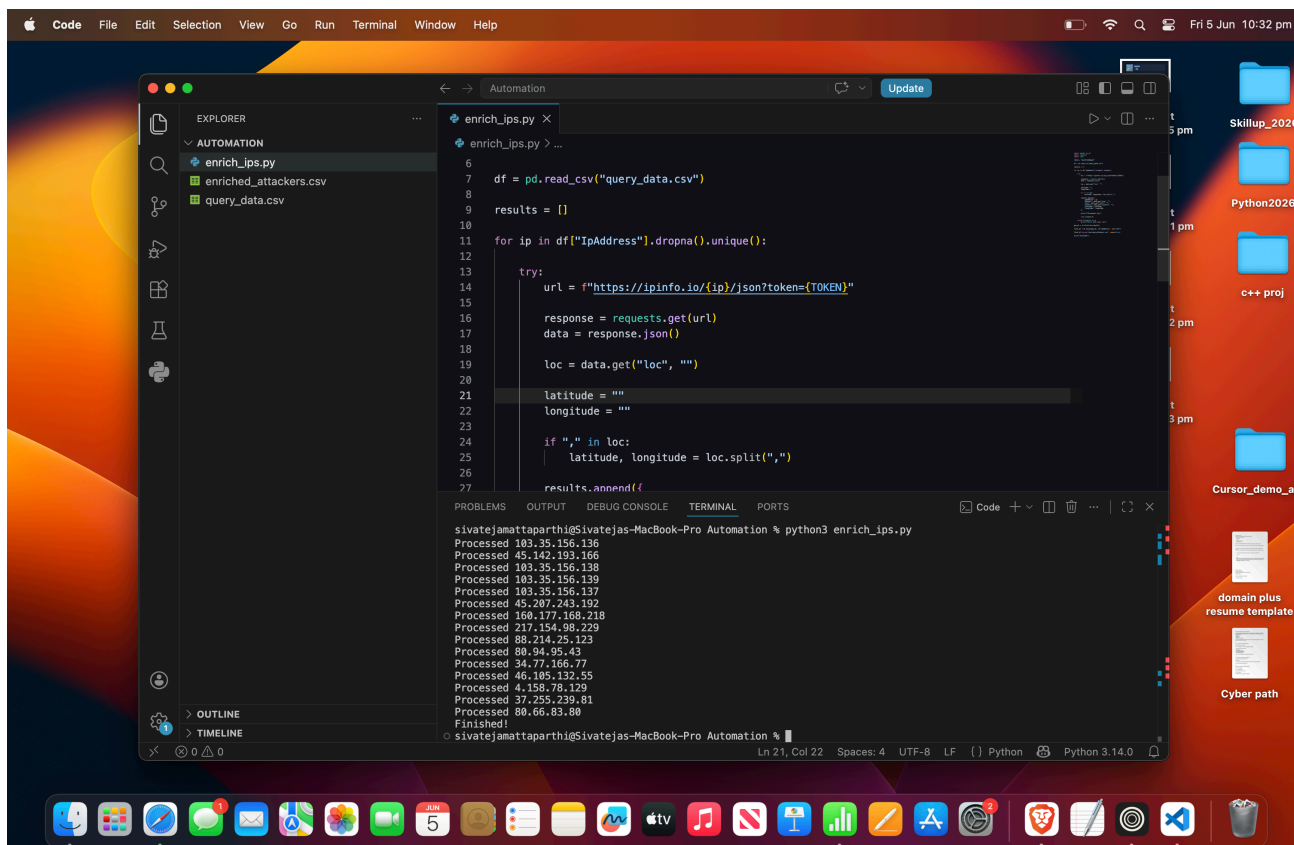
Data Preparation for Sentinel Watchlists

During the process of importing the enriched dataset into Microsoft Sentinel Watchlists, ‘an ingestion limitation was encountered.’

The original CSV contained username-related fields extracted from the authentication logs. However, due to Sentinel Watchlist import requirements and field limitations, these columns were not required for geolocation-based visualisation and were therefore removed **manually** from the dataset prior to import.

The final watchlist dataset focused exclusively on attack source information, including:

- IP Address
- Attack Count
- Network Provider
- Geographic Coordinates
- City
- Country



7) Sentinel Watchlist Creation and Attack Visualisation

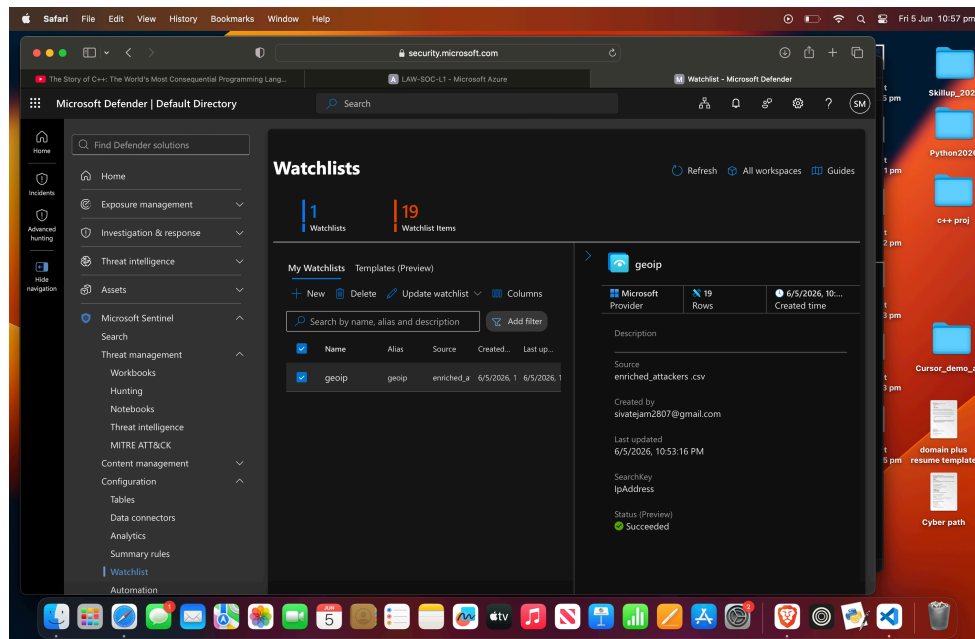
After enriching the attacker IP addresses with geolocation and network ownership information, the resulting dataset was prepared for import into Microsoft Sentinel. This dataset was imported into Microsoft Sentinel as a custom Watchlist named geoiip.

During the Watchlist creation process, the “IP Address” field was selected as the Search Key. This allowed Sentinel to efficiently reference and correlate attacker information using IP addresses obtained from security logs.

Following the import process, the Watchlist was verified using KQL queries to ensure that all records were successfully ingested.

The following query was used:

```
_GetWatchlist('geoiip')
```



Workbook Creation

To visualise attack activity geographically, a custom Microsoft Sentinel Workbook was created.

The Workbook provided an interactive dashboard capable of displaying attacker locations on a world map using the latitude and longitude values contained within the GeoIP Watchlist.

A new Workbook was created within Microsoft Sentinel and configured to retrieve attack information from both the SecurityEvent table and the GeoIP Watchlist.

To customise the visualisation, the Workbook's Advanced Editor was used to import a JSON configuration defining the map settings.

The configuration specified:

- Latitude and Longitude mapping
- Marker sizing based on attack volume
- Geographic labels
- Heatmap styling
- Attack count aggregation

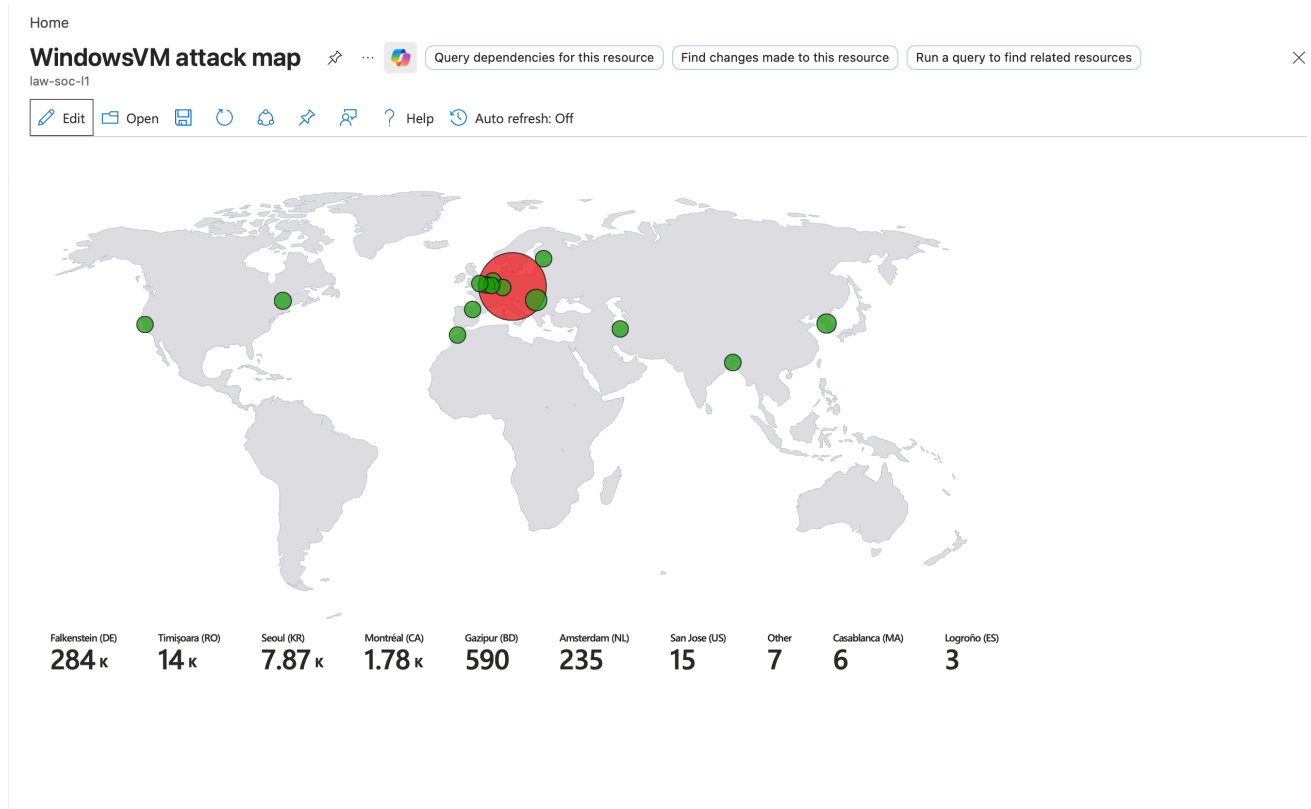
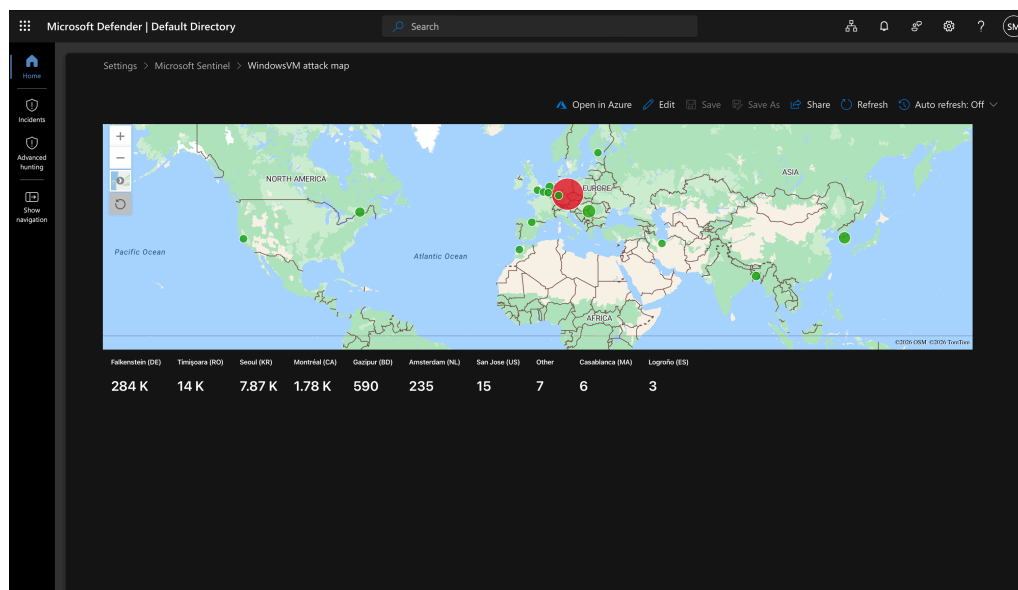
This configuration transformed raw security data into an interactive geographic visualisation capable of highlighting attack hotspots around the world.

Final Attack Map

After completing the Workbook configuration, attack activity was successfully visualised on a global map.

Each marker represented an attacker location, while marker size reflected the volume of failed authentication attempts originating from that source. This provided a clear representation of the geographic distribution of attack activity targeting the internet-exposed virtual machine.

The visualisation revealed attack attempts originating from multiple countries and demonstrated how quickly publicly accessible systems attract automated scanning and credential-based attacks.



8) Findings, Lessons Learned and Conclusion

Findings

During the monitoring period, the internet-exposed Windows virtual machine attracted a significant amount of unsolicited attack activity from around the world. Analysis of Windows Security Event Logs revealed thousands of failed authentication attempts originating from multiple countries and network providers.

A total of **19 unique attacker IP addresses** were identified and enriched with geolocation information. One IP address alone generated more than **284,000 failed login attempts**, potentially involving a bot conducting large-scale brute-force password guessing attempts against the publicly accessible system. highlighting the scale and persistence of automated brute-force and credential-based attacks targeting publicly accessible systems.

The attack map visualisation demonstrated that malicious activity was geographically distributed across several regions, including Europe, Asia, North America, and the Middle East. The results showed how quickly internet-facing systems become targets for automated scanning and attack infrastructure

Lessons Learned:

This project provided valuable hands-on experience with cloud security monitoring and Security Operations Center (SOC) workflows. The project also demonstrated the importance of proper security controls, such as restrictive firewall rules, strong authentication mechanisms, and continuous monitoring of security events.

Key lessons learnt:

- Deploying and managing cloud resources within Microsoft Azure.
- Configuring Microsoft Sentinel as a cloud-native SIEM platform.
- Collecting and analysing Windows Security Event Logs.
- Using Kusto Query Language (KQL) to investigate security events.
- Automating threat intelligence enrichment using Python and external APIs.
- Creating and managing Microsoft Sentinel Watchlists.
- Building interactive dashboards and geospatial attack visualisations.
- Understanding how exposed systems are targeted by automated attack tools on the public internet.

Conclusion:

The project provided practical experience with cloud security technologies and highlighted the value of centralised logging, threat intelligence, and security analytics in understanding attacker behaviour. Overall, the implementation served as an effective introduction to SOC operations and demonstrated how security professionals can transform raw log data into meaningful and actionable insights.

This project reinforced the importance of continuous monitoring, threat intelligence enrichment, and proactive security analysis in defending modern cloud environments.

Appendix A - Python Enrichment Script

```
import pandas as pd

import requests

import time

TOKEN = "23dfsafafk222" #Not real token, replace this with your own token.

df = pd.read_csv("query_data.csv")

results = []

for ip in df["IpAddress"].dropna().unique():

    try:

        url = f"https://ipinfo.io/{ip}/json?token={TOKEN}"

        response = requests.get(url)

        data = response.json()

        loc = data.get("loc", "")
```

```
latitude = ""

longitude = ""

if "," in loc:

    latitude, longitude = loc.split(",")

results.append({

    "IpAddress": ip,

    "Network": data.get("org", ""),

    "City": data.get("city", ""),

    "Country": data.get("country", ""),

    "Latitude": latitude,

    "Longitude": longitude

})

print(f"Processed {ip}")

time.sleep(0.5)

except Exception as e:

    print(f"Error with {ip}: {e}")

geo_df = pd.DataFrame(results)

final_df = df.merge(geo_df, on="IpAddress", how="left")

final_df.to_csv("enriched_attackers.csv", index=False)

print("Finished!")
```

Appendix B - KQL Queries

1) SecurityEvent

| where EventID == 4625

2) SecurityEvent

| where EventID == 4625

| project TimeGenerated, Account, Computer, EventID, Activity, IPAddress

3) SecurityEvent

| where EventID == 4625

| summarize Attempts=count() by IPAddress

| order by Attempts desc

4) SecurityEvent

| where EventID == 4625

| summarize Attempts=count() by IPAddress, Account

| order by Attempts desc

5) _GetWatchlist('geoip')

Appendix C - JSON Script (Map Plotting)

```
{
  "type": 3,
  "content": {
    "version": "KqlItem/1.0",
    "query": "let GeoIP = _GetWatchlist('geoip');\nSecurityEvent\n| where EventID == 4625\n| summarize FailureCount=count() by IpAddress\n| join kind=inner GeoIP on IpAddress\n| project FailureCount, AttackerIp=IpAddress, Latitude, Longitude, City, Country, friendly_location=strcat(City,'(',Country,')')",
    "size": 3,
    "timeContext": {
      "durationMs": 2592000000
    },
    "queryType": 0,
    "resourceType": "microsoft.operationalinsights/workspaces",
    "visualization": "map",
    "mapSettings": {
      "locInfo": "LatLong",
      "latitude": "Latitude",
      "longitude": "Longitude",
      "sizeSettings": "FailureCount",
      "sizeAggregation": "Sum",
      "opacity": 0.8,
      "labelSettings": "friendly_location",
      "legendMetric": "FailureCount",
      "legendAggregation": "Sum",
      "itemColorSettings": {
        "nodeColorField": "FailureCount",
        "colorAggregation": "Sum",
        "type": "heatmap",
        "heatmapPalette": "greenRed"
      }
    }
  },
  "name": "Global RDP Attack Map"
}
```